Note 2: Operator Factorization and the Structural Obstruction

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1 Recap

From Note 1: the Apéry generating function $F(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3\binom{2n}{n}} x^n$ satisfies:

$$x^2(4+x)\,F''' + x(10+3x)\,F'' + (2+x)\,F' = 1 \tag{1}$$

with $F(0) = 0$, $F'(0) = \frac{1}{2}$, $F''(0) = -\frac{1}{24}$, and $F(1) = \frac{2}{5}\zeta(3)$.

The substitution $x = 1-e^{-t}$ converts this to an IVP on $[0,\infty)$ with rational initial conditions but a non-polynomial right-hand side (due to $1/[x^2(4+x)]$ in the $\dot{y}_2$ equation).

Time reparametrization $d\tau/dt = 1/[x^2(4+x)]$ makes the RHS polynomial but turns $x = 0$ into a fixed point.

2 Operator Factorization

Theorem 1 (Factorization)**.** *The differential operator* $L = x^2(4+x)D^3 + x(10+3x)D^2 + (2+x)D$ *factors as*

$$L = D \circ M_1 \circ M_2$$

where $M_2 = xD + 1$ *and* $M_1 = x(4+x)D + (2+x)$*.*

Proof. Setting $H = F'$, the equation $L[F] = 1$ becomes $M[H] = 1$ where $M = x^2(4+x)D^2 + x(10+3x)D + (2+x)$.

We seek α such that $M = M_1 \circ M_2$ with $M_2[H] = xH' + \alpha H$ and $M_1[G] = c(x)G' + d(x)G$. Expanding $M_1 \circ M_2$ and matching coefficients:

$$\begin{aligned} c(x)\cdot x &= x^2(4+x) \quad \Rightarrow \quad c(x) = x(4+x),\\ d(x)\cdot\alpha &= 2+x. \end{aligned}$$

The remaining coefficient equation yields $(\alpha-1)^2 = 0$, so $\alpha = 1$, giving $d(x) = 2+x$.

Verification: $M_1[M_2[H]] = x(4+x)(2H' + xH'') + (2+x)(xH' + H) = x^2(4+x)H'' + x(10+3x)H' + (2+x)H$. □ □

Proposition 1 (First-order ODE for G)**.** *Define* $G = M_2[F'] = xF'' + F'$*. Then* G *satisfies the first-order linear ODE*

$$x(4+x)\,G' + (2+x)\,G = 1 \tag{2}$$

with $G(0) = \frac{1}{2}$*. The leading coefficient* $x(4+x)$ *has a* ***simple*** *zero at* $x = 0$ *(vs. the double zero* $x^2(4+x)$ *in the original ODE).*

Proposition 2 (Series for G)**.** *The Taylor coefficients of* $G(x) = \sum_{n=0}^{\infty} G_n x^n$ *are*

$$G_n = \frac{(-1)^n}{2(2n+1)\binom{2n}{n}}, \qquad G_0 = \tfrac{1}{2},$$

satisfying the recurrence $G_n = -\frac{n}{2(2n+1)}\,G_{n-1}$*.*

3 The Indicial Structure

Proposition 3 (Indicial equation)**.** *The indicial polynomial of the original ODE* (**??**) *at* $x = 0$ *is*

$$I(\rho) = 2\rho^2(2\rho - 1),$$

with roots $\rho = 0$ *(multiplicity 2) and* $\rho = \frac{1}{2}$ *(multiplicity 1).*

The double root at $\rho = 0$ is the source of the structural obstruction.

4 Five-Variable Polynomial System

Using the factorization and reparametrizing time by $d\tau/dt = 1/x$, with auxiliary variable $w = 1/(4 + x)$:

$$\begin{aligned} \dot{F} &= H\,x(1-x), & F(0) &= 0, \\ \dot{H} &= (G-H)(1-x), & H(0) &= \tfrac{1}{2}, \\ \dot{G} &= (1-x)\,w\big[1-(2+x)G\big], & G(0) &= \tfrac{1}{2}, \\ \dot{w} &= -w^2\,x(1-x), & w(0) &= \tfrac{1}{4}, \\ \dot{x} &= x(1-x), & x(0) &= 0. \end{aligned} \tag{3}$$

Observation 1. *System* (**??**) *has:*

1. *Polynomial right-hand side (degree* ≤ 3*).*
2. *All rational initial conditions.*
3. *Bounded trajectories (verified numerically from* $x = \varepsilon > 0$*).*

However, $x = 0$ *is a fixed point of* $\dot{x} = x(1-x)$*, so the trajectory is frozen at the initial conditions for all* τ*.*

5 The Blow-Up Chain and Infinite Regression

5.1 First blow-up

In the original t-domain (where $\dot{x} = 1 - x$ has no fixed point), the factored system has:

$$\dot{H} = \frac{(G-H)(1-x)}{x}.$$

Since $G(0) = H(0) = \frac{1}{2}$, define $P = (G - H)/x$. Numerically $P(0) = -\frac{1}{24} = F''(0)$.

5.2 Equation for P

The ODE for P in t-time involves

$$\dot{P} = \frac{(1-x)\big[1-(2+x)H - x(10+3x)P\big]}{x(4+x)}.$$

Define the numerator $N_1 = 1 - (2 + x)H - x(10 + 3x)P$. Then:

- $N_1(0) = 1 - 2 \cdot \frac{1}{2} = 0$.
- $N_1'(0) = 0$ (verified numerically and algebraically).
- $N_1 \sim x^2$ as $x \to 0$, with $\lim_{x\to 0} N_1/x^2 = \frac{2}{45}$.

5.3 Second blow-up and beyond

To absorb the $1/x$ in $\dot{P}$, define $S_2 = N_1/x^2$ (finite at $x = 0$). But the ODE for S_2 again involves a division by x, requiring a third blow-up variable, and so on indefinitely.

Theorem 2 (Infinite regression)**.** *The blow-up chain for the factored Apéry system does not terminate in finitely many steps. At each level k, the new variable V_k (absorbing $1/x$ from level $k-1$) satisfies an ODE whose right-hand side contains a new $0/0$ ratio at $x = 0$, requiring a further blow-up at level $k+1$.*

This infinite regression is a consequence of the double indicial root $\rho = 0$ (multiplicity 2) in the original ODE.

Remark 1. For an ODE with only *simple* indicial roots at a regular singular point, a single blow-up suffices to resolve the singularity. The double root creates a logarithmic solution $F_{\log}(x) = F_1(x)\log(x) + \tilde{F}(x)$, and the logarithmic monodromy is what prevents finite resolution.

6 Summary of Obstruction

Approach	**Polynomial RHS?**	**Computes $\zeta(3)$?**
Original t-domain	No ($1/x^2$)	Yes
Time reparam. by $x^2(4+x)$	Yes	No (fixed point)
Factored + reparam. by x	Yes	No (fixed point)
$\tau e^{-\tau}$ kick	Yes	No (breaks chain rule)
Blow-up chain	Approaches polynomial	Infinite steps needed

The core obstruction: the Apéry generating function ODE has a regular singular point at $x = 0$ with a double indicial root. This singularity cannot be resolved into a polynomial PIVP by any finite algebraic manipulation.

7 What Would Work

A generating function or series for $\zeta(3)$ whose ODE has:

1. No singular points in $[0, 1]$, **or**
2. Only regular singular points with *simple* (multiplicity 1) indicial roots, **or**
3. A singular point where the indicial roots are non-negative integers with multiplicity 1 (apparent singularity, removable).

Finding such an ODE is the next direction.